

CO₂ Dashboard Using R Shiny: Exploratory Analysis of Global CO₂ emissions

**Project Report
Course : MTH208**

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1 Introduction and Background

Climate change is one of the most urgent challenges of the 21st century. Among the greenhouse gases responsible for global warming, carbon dioxide remains the primitive contributor. Understanding the various trends and disparities in CO₂ emissions across countries is key for designing effective environmental and climate policies. This report uses a dataset scraped from Worldometer (site: <https://www.worldometers.info/co2-emissions/>) and an R Shiny dashboard to explore these patterns.

2 Objectives

- **Primary objective:** Explore global and cross-country CO₂ emission patterns and provide interactive tools for inspection.
- **Specific objectives:**
 1. Document the data source and variables.
 2. Explain the main visualizations (top emitters, population vs emissions, density of per-capita emissions, bubble chart, pie chart of share of emissions).
 3. Describe the R Shiny dashboard UI and how it supports interactive exploration.
 4. Summarize key findings and discuss how the app improves interpretability.



3 Data: sources and description

3.1 Source

The primary source is the Worldometer CO₂ emissions page (scraped HTML table).

3.2 Key variables

Table 1: Key variables (as derived in the cleaning step)

Variable	Description
country	Country name (cleaned)
emissions_tons	Total CO ₂ emissions (tons)
population_2022	Population of the country (2022)
per_capita_tons	CO ₂ per person (tons/person)
share_percent	Country share of global CO ₂ (%)
change_percent	Reported change percentage from past year
emissions_per_million	Amount of emissions per every million people

4 Data collection and cleaning

4.1 Web scraping

The data is scraped using the **rvest** package. A compact extraction snippet:

```
library(rvest)
url <- "https://www.worldometers.info/co2-emissions/"
page <- read_html(url)
table_node <- page %>% html_node("table")
co2_raw <- table_node %>% html_table(fill = TRUE)
```

Notes: the script reads the first table node and converts it to a data frame. Always check that the table structure has not changed on the source site and comply with the website's terms.

4.2 Cleaning

R packages such as **dplyr** were used to rename columns for clarity, remove special characters, handle missing values, and create derived variables. Representative cleaning function from the script:

```
clean_num <- function(x) {
  x <- gsub(",", "", x)
  x <- gsub("%", "", x)
  x <- gsub(" ", "-", x)
  x <- gsub("[^0-9\\.]", "", x)
  suppressWarnings(as.numeric(x))
}
```

Importance of data cleaning

Data cleaning is one of the most important steps before analyzing or visualizing any dataset. In real life, data often comes with mistakes — missing values, extra spaces, inconsistent character or negative entries, or even wrong entries. This could be due to faulty data storage, or calculation errors or approximations. If we avoid data cleaning, these issues can lead to wrong insights or confusing results.

5 Libraries

The R script uses the following primary packages:

- `rvest`, `httr` — web scraping and HTTP handling.
- `dplyr` — data cleaning.
- `ggplot2` — plotting.
- `shiny` — interactive dashboard.

6 Exploratory Data Analysis

6.1 Key research questions that arise:

Which countries are the largest contributors to global CO₂ emissions?
 How concentrated are global CO₂ emissions among the top 10 emitters?
 Which countries have the highest and lowest per-capita CO₂ emissions?
 What is the overall distribution of CO₂ emissions across all countries?
 What is the correlation between population and total CO₂ emissions?

7 Key Findings

7.1 Top Emitters (Bar chart)

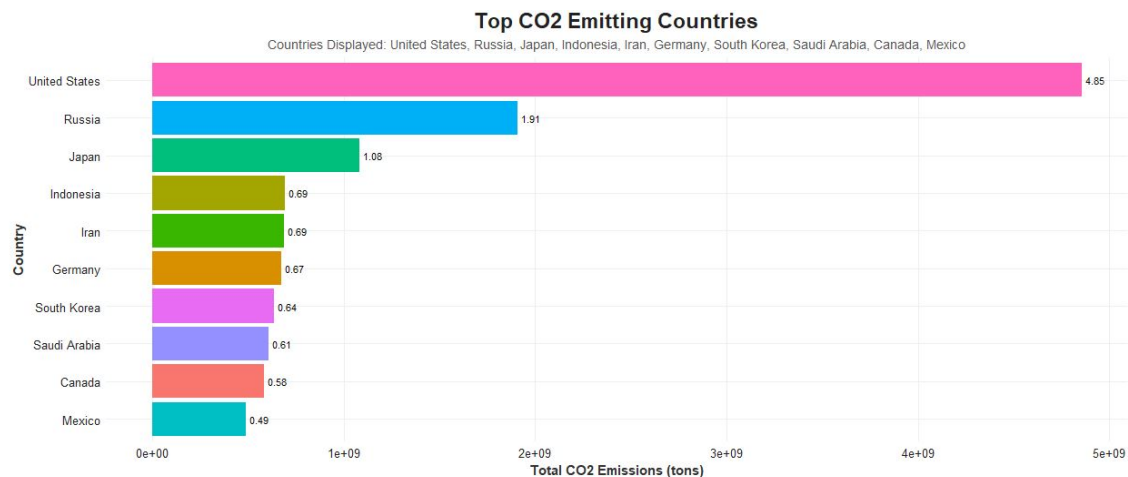


Figure 1: Top CO₂ emitting countries (tons).

Discussion: The bar chart ranks countries by absolute emissions. Note how a small set of countries dominates total global emissions; we discuss the top N shown in our app and their absolute values.

For the graph below, pertaining to a section of countries with comparatively lesser population, we see that these countries have smaller populations but very high industrial activity and energy consumption per person, resulting in high amount of total emissions. Their emission levels typically lie between 0.3 to 0.6 billion tons, indicating that even with lesser populations, economic development and energy dependence play major roles in driving emissions upward.

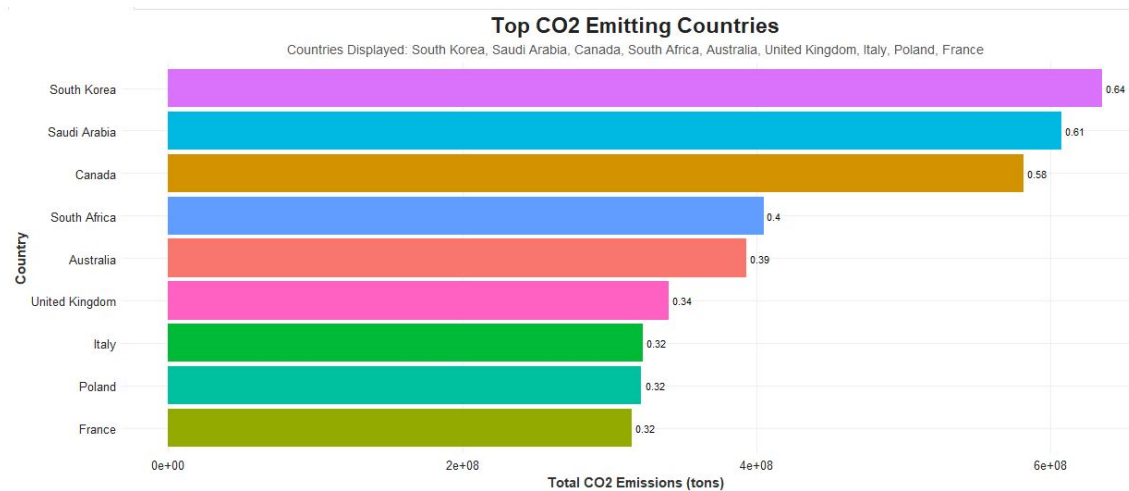


Figure 2: Top CO₂ emitting countries among lesser populated countries

7.2 Emissions vs Population (Scatter)

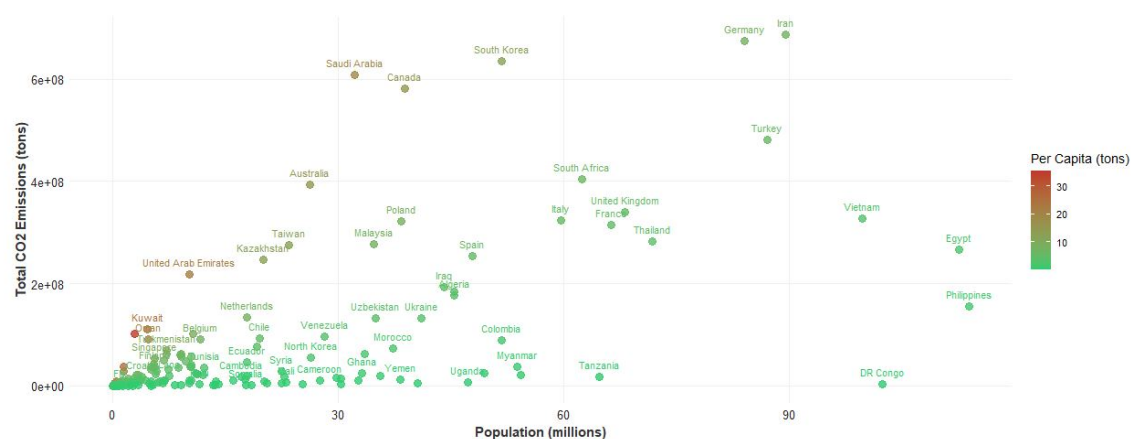


Figure 3: Total emissions vs population, colored by per-capita emissions.

Discussion: This scatter plot illustrates the relationship between a country's population size (in millions) and its total CO₂ emissions (in tons), with the color gradient representing per-capita emissions (tons per person). Overall, the pattern shows a positive correlation between population and total CO₂ emissions — countries with larger populations tend to emit more overall. However, the color shading reveals that population alone does not explain emission intensity. Some countries with moderate populations, such as Saudi Arabia, Canada, and Australia, exhibit very high per-capita emissions (shown in brownish hues), indicating high carbon output per person. In contrast, nations like India, Vietnam, and the Philippines have large populations but relatively low per-capita emissions (green shades), meaning their per-person contribution is small despite high total emissions. The plot thus underscores how both population scale and economic/industrial intensity shape total CO₂ output, highlighting disparities between developed and developing economies in carbon intensity.

7.3 Per-capita Distribution (Density)

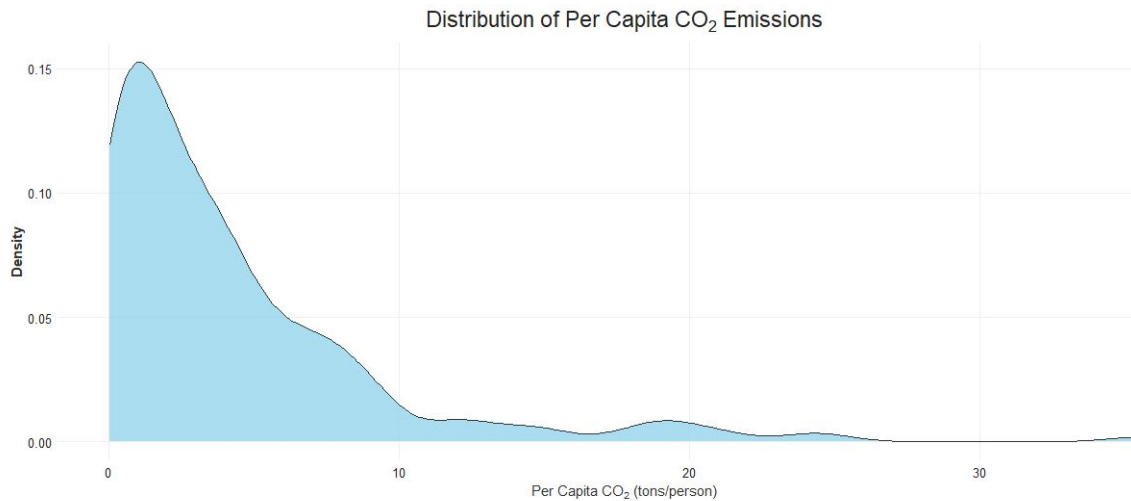


Figure 4: Distribution of per-capita CO₂ emissions across countries.

Discussion: This density plot illustrates the distribution of per capita CO₂ emissions (in tons per person) across countries. The distribution is highly right-skewed, with the majority of nations emitting relatively low levels of CO₂ per person—mostly under 5 tons per capita. The tall peak near zero indicates that a large number of developing or less industrialized countries contribute minimally on a per-person basis. However, the long right tail shows that a smaller group of countries have exceptionally high per capita emissions, often exceeding 15–20 tons, typical of highly industrialized or resource-rich economies. This skewed pattern highlights the global inequality in carbon emissions, where a few nations bear a disproportionately large share of individual emission footprints.

8 Shiny App: interface and user flow

The Shiny UI has the following structure:

- **Sidebar controls:** sliders for `top_n` (Top N countries), population range (millions), per-capita range, total emissions range (in billions).
- **Top metrics row:** total emissions, mean per-capita, median emissions, correlation (emissions vs population).
- **Tabs:** Top Emitters, Scatter, Density, CO₂ Share Pie Chart, Key Findings, About.

The ‘Key Findings’ tab in the app computes summary metrics.

```
---- Summary of Key Metrics ----
Countries considered: 205
Average per capita emissions: 4.32
Average global share (%): 0.47
Variance in emissions: 9.521023e+17
Correlation (CO2 vs Pop): 0.809
```

8.1 Dashboard screenshot

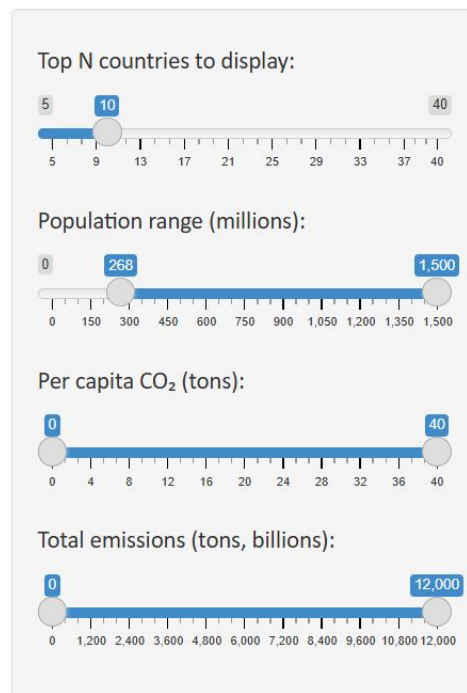


Figure 5: Screenshot of the sliders and filters panel



Figure 6: Screenshot of the center panel

Interpretation :

- High correlation indicates population is a strong predictor of total emissions, but per-capita shading shows differences in intensity of emissions per person.
- Concentration among top emitters indicates targeted policy in major emitting countries can yield large reductions.
- Per-capita outliers may reflect small, wealthy countries with high industrial activity or energy-intense economies.

9 Limitations

While this project provides valuable insights into global CO₂ emission patterns, it has several drawbacks. The analysis is based solely on data scraped from Worldometers, which restricts access to more granular or historical datasets. Due to this, deeper econometric relationships (such as emissions versus GDP or energy consumption) could not be fully explored. Also, the app intentionally avoids the use of advanced visualization or UI libraries—such as `bslib` or external CSS frameworks—to maintain simplicity and compatibility. This restriction, however, limits the overall aesthetic customization and interactivity of the interface.

10 Acknowledgement

We would like to express our sincere thanks to Dr. Akash Anand for providing an opportunity to work on this project. His mentoring and advice helped a lot during the course, which was essential for us to smoothly execute the project. Also, thanks to our team for the sincere efforts throughout the whole work period.

11 Ethics and Feasibility

Sources are properly cited within the app. Analysis has been presented objectively with an aim to educate and inform.

12 Reproducibility Notes

1. Install the required R packages: `shiny`, `ggplot2`, `dplyr`, `rvest`
2. Submitted files:
 - a. **Project Report(pdf)**: A comprehensive report about the entire project
 - b. **Cleaned dataset(csv)**: Dataset obtained after cleaning the scraped data.
 - c. **Scraping Code(R script)**: This code scrapes out the data from the worldometers website, cleans it, and stores (by `write.csv()` at the end) the cleaned dataset mentioned above.
 - d. **App Code(R Script)**: This is the code for the shiny app. At the beginning of the app, the code reads the same dataset that was stored at the end of the previous code.

It is requested to adjust/set the working directory properly in case the submitted dataset is downloaded to use the shiny app code directly.

13 References

Worldometer CO₂ emissions 2022 data: <https://www.worldometers.info/co2-emissions/>